

SEUNGHUN BAEK

 [Personal Page](#)  [Google Scholar](#)

#4303 RIST Building IV, POSTECH, 77 Cheongam-ro, Nam-gu, Pohang, South Korea

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RESEARCH INTERESTS

My research focuses on **robust representation learning** under practical data constraints, including **limited samples**, **class imbalance**, and **incomplete multimodal observations**. I aim to translate these methods into reliable real-world application, particularly in medical settings involving neurodegenerative diseases.

EDUCATION

POSTECH

Integrated M.S. and Ph.D. in Artificial Intelligence

Pohang, South Korea

Aug 2021 - Feb 2027 (Expected)

- Supervised by [Prof. Won Hwa Kim](#) at [Medical Information Processing Lab.](#)

Carnegie Mellon University

Visiting Student in Language Technology Institute

Pittsburgh, United States

Jan 2022 - Jul 2022

Korea National University of Arts

Exchange Student

Seoul, South Korea

Jun 2019 - Aug 2019

POSTECH

B.S. in Computer Science and Engineering

Pohang, South Korea

March 2017 - Aug 2021

PUBLICATIONS (*: EQUAL CONTRIBUTION)

- [1] *Residual-Guided Expert Specialization for Incomplete Multimodal Learning*
Seunghun Baek, Jihwan Park, Jaeyoon Sim, Minjae Jeong, Hoseok Lee, Won Hwa Kim
European Conference on Computer Vision (ECCV), 2026
- [2] *Set-Inclusive Uncertainty Modeling for Robust Brain Tumor Segmentation*
Seunghun Baek*, Jihwan Park*, Jaeyoon Sim, Hoseok Lee, Seungjoo Lee, Won Hwa Kim
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026
- [3] *ENC-ODE: Event-level Neurodegenerative Modeling in Continuous Time with Neural ODEs*
Yujee Song*, **Seunghun Baek***, Guorong Wu, Won Hwa Kim
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026
- [4] *Spectral-Temporal State Space Modeling on Functional Brain Networks*
Jaeyoon Sim*, Hoseok Lee*, Jihwan Park, **Seunghun Baek**, Won Hwa Kim
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026
Provisional Accept (Top ~9%)
- [5] *Learning Multi-Scale Hypergraph for High-Order Brain Connectivity Analysis*
Jaeyoon Sim, Soojin Hwang, **Seunghun Baek**, Guorong Wu, Won Hwa Kim
International Conference on Machine Learning (ICML), 2026
- [6] *Mark4D: Temporally-Consistent Watermarking for 4D Gaussian Splatting*
Jaejin Lee*, Minjae Jeong*, Joonhyuk Park, Yechan Hwang, **Seunghun Baek**, Won Hwa Kim
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025
- [7] *MNM: Multi-level Neuroimaging Meta-analysis with Hyperbolic Brain-Text Representations*
Seunghun Baek, Jaejin Lee, Jaeyoon Sim, Minjae Jeong, Won Hwa Kim
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2025
Provisional Accept (Top ~9%)

- [8] *OCL: Ordinal Contrastive Learning for Imputating Features with Progressive Labels*
Seunghun Baek*, Jaeyoon Sim*, Guorong Wu, Won Hwa Kim
 Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
Provisional Accept (Top ~11%)
- [9] *Modality-Agnostic Style Transfer for Holistic Feature Imputation*
Seunghun Baek*, Jaeyoon Sim*, Mustafa Dere, Minjeong Kim, Guorong Wu, Won Hwa Kim
 IEEE International Symposium on Biomedical Imaging (ISBI), 2024
Oral presentation (Top ~14%)
- [10] *Learning Covariance-Based Multi-Scale Representation of Neuroimaging Measures for Alzheimer Classification*
Seunghun Baek, Injun Choi, Mustafa Dere, Minjeong Kim, Guorong Wu, Won Hwa Kim
 IEEE International Symposium on Biomedical Imaging (ISBI), 2023

PROJECT

AI-Driven BGA Solder Joint Failure Detection of PCB Assembly

Mar 2023 - Oct 2023

Granted by SK Hynix

- Related Topic: Long-tailed Recognition. Post-localization Refinement.
- Developed a soldering defect inspection system that reduces inspection time and operator-dependent variability.
- Selected as the **2nd Best Patent** among 27 industry-academia collaborative patents at SK Hynix. [Press Release](#)

PATENTS

- [1] *Ordinal Contrastive Learning for Imputating Features with Progressive Labels*
Seunghun Baek, Jaeyoon Sim, Won Hwa Kim
 US Patent App. 19/336,903, 2025
- [2] *Future Event Prediction Method using Decoupled Marked Temporal Point Process and Analysis Apparatus*
 Yujee Song, Donghyun Lee, **Seunghun Baek**, Won Hwa Kim
 US Patent App. 19/388,012, 2025
- [3] *Neuroimaging Meta-Analysis Method based on Hyperbolic Brain-Text Representations*
Seunghun Baek, Jaejin Lee, Jaeyoon Sim, Won Hwa Kim
 KR Patent App. 10-2025-0134582
- [4] *Future Event Prediction Method using Decoupled Marked Temporal Point Process and Analysis Apparatus*
 Yujee Song, Donghyun Lee, **Seunghun Baek**, Won Hwa Kim
 KR Patent App. 10-2025-0108586
- [5] *Ordinal Contrastive Learning for Imputating Features with Progressive Labels*
Seunghun Baek, Jaeyoon Sim, Won Hwa Kim
 KR Patent App. 10-2024-0128729
- [6] *Solder Paste Inspection Device using Neural Network and Operation Method Thereof*
Seunghun Baek, Jaeyoon Sim, Won Hwa Kim
 KR Patent App. 10-2024-0054209

HONORS & AWARDS

Best Paper Award (Bronze Prize)

Feb 2026

38th Workshop on Image Processing and Image Understanding (IPIU)

Global Core Talent Development Program (AI Intensive Program at CMU)

2022

Institute of Information & Communications Technology Planning & Evaluation (IITP)

National Science & Technology Scholarship

2019

Korea Student Aid Foundation

PROFESSIONAL SERVICE

Reviewer, Medical Image Computing and Computer Assisted Intervention (MICCAI): 2026

TEACHING EXPERIENCES

<i>(Teaching Assistant) AI & Big Data Academy at POSCO</i>	Fall 2024
<i>(Teaching Assistant) CSED211: Introduction to Computer Software Systems</i>	Fall 2023
<i>(Teaching Assistant) CSED490F: Signal Processing</i>	Fall 2022

EXTRA ACTIVITIES

<i>Residential Advisor</i>	2019-2020
<i>Student Mentoring Program - (CSED101: Programming & Problem solving)</i>	2019-2020
<i>Research Intern (Korea Institute of Science and Technology)</i>	Jun 2019 - Jul 2020
<i>Student Mentor at 9th Einstein Class (Hosted by Korea Hydro & Nuclear Power Co.)</i>	Dec 2017 - Feb 2018
<i>Software Education Student Volunteer Corps (Hosted by NAVER Connect Foundation)</i>	Aug 2017 - Dec 2017